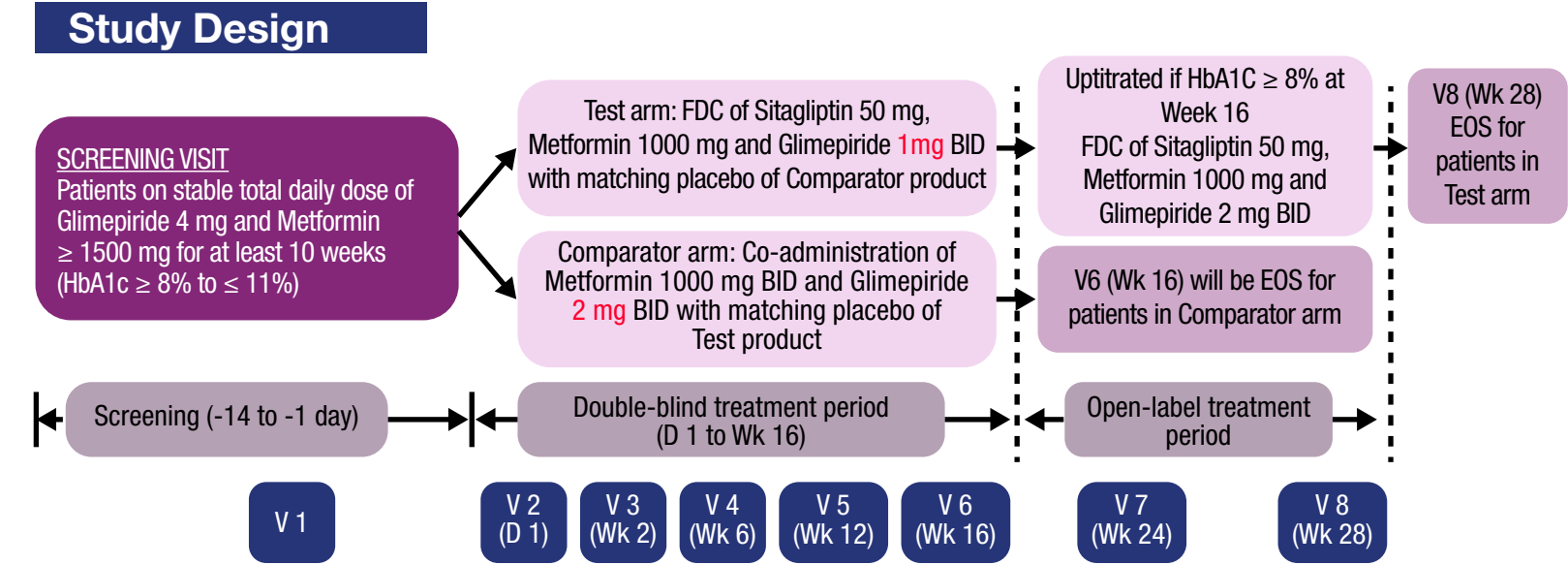


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- BACKGROUND
- Cross-sectional data from the CURES suggests that Indians consume high amounts of refined grains (~47% of total calories), which is associated with significant increases in FBG ($p=0.007$).¹ In a recent meta-analysis, admission FBG was found to be an independent predictor of disease severity, with every one mmol/L increase in FBG translating into a 33% increased risk of disease severity. Thus, FBG control is essential to reduce the severity of disease.²
 - As per RSSDI guidelines 2022, target values for glucose control in non-pregnant adults is HbA1c < 7.0%, FBG < 115 mg/dL and PPBG < 160 mg/dL.³
 - Thus, we conducted this post-hoc analysis of a phase 3 study to evaluate the improvement in FBG at Week 16.

- METHODS
- This is a post-hoc analysis of a phase 3, randomized, active-controlled, double-blind, double-dummy study (16 Weeks) followed by an open-label, single-arm study (12 Weeks) [CTRI/2021/11/038169].
 - Study included adult Indian patients with HbA1c $\geq 8\%$ and $\leq 11\%$ taking Glimepiride 4 mg and Metformin ≥ 1500 mg/day for at least 10 weeks prior to screening. Patients received FDC of Sitagliptin+Glimepiride+Metformin tablet (50+1+1000 mg) twice daily (BID) (Test arm) or co-administration of Metformin 1000 mg BID and Glimepiride 2 mg BID (Comparator arm) for 16 Weeks

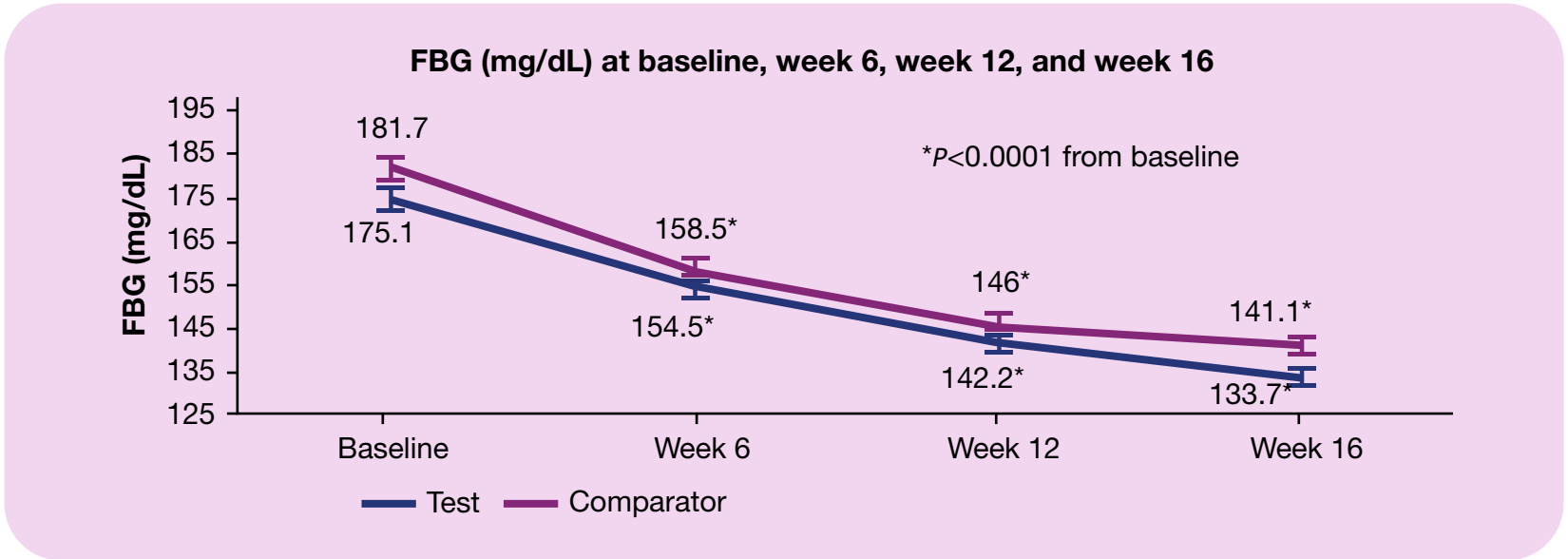


- In this post hoc analysis, we evaluated the improvement in FBG at Week 16.

- RESULTS
- Baseline Demographics
- A total of 392 (190 in Test and 202 in Comparator) patients were included in this post hoc analysis.
 - Baseline characteristics were comparable between Test arm and Comparator arm.

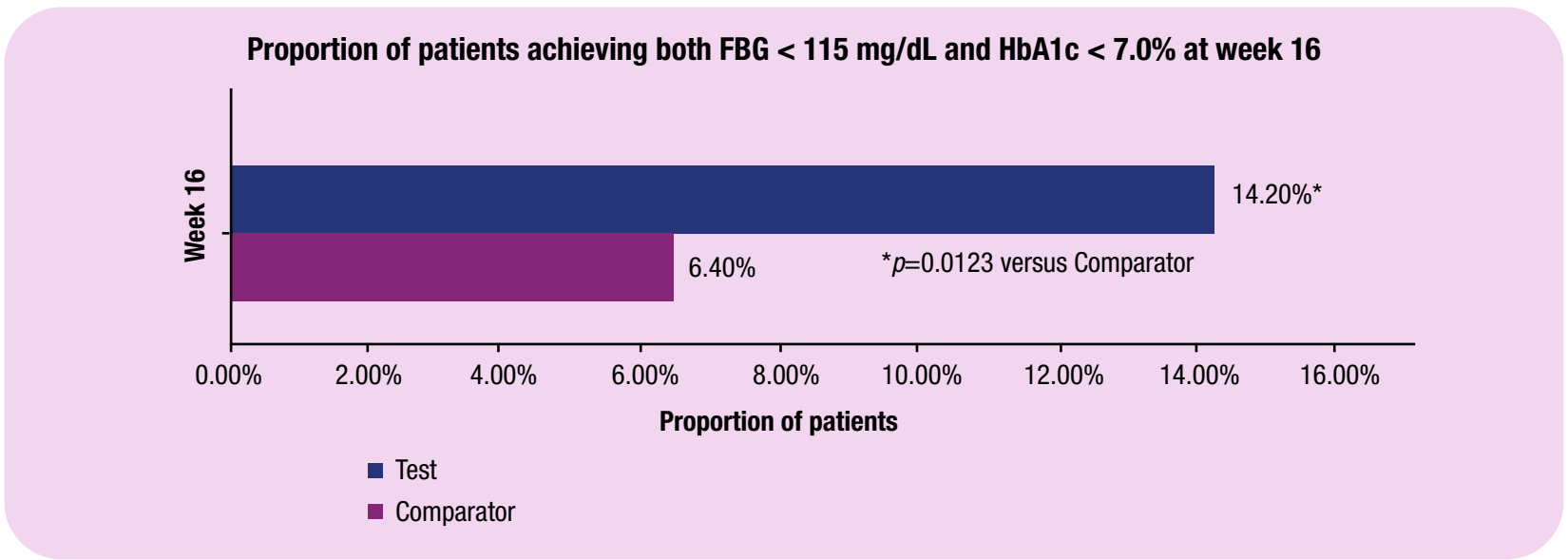
	Test arm N=190	Comparator arm N=202	Overall N=392
Gender	n (%)		
Male	102 (53.7%)	121 (59.9%)	223 (56.9%)
Female	88 (46.3%)	81 (40.1%)	169 (43.1%)
Mean $\pm$ SD			
Age (years)	47.57 $\pm$ 8.57	47.01 $\pm$ 9.16	47.29 $\pm$ 8.87
Height (cm)	162.8 $\pm$ 7.68	163.5 $\pm$ 8.87	163.2 $\pm$ 8.31
Weight (kg)	67.51 $\pm$ 9.93	69.16 $\pm$ 10.34	68.36 $\pm$ 10.16
Body Mass Index (kg/m ²)	25.49 $\pm$ 3.61	25.95 $\pm$ 3.87	25.73 $\pm$ 3.75
Glycated hemoglobin (%)	9.18 $\pm$ 0.75	9.14 $\pm$ 0.74	9.16 $\pm$ 0.75
Fasting Blood Glucose (mg/dL)	177.2 $\pm$ 38.31	183.9 $\pm$ 40.30	180.6 $\pm$ 39.44
Postprandial Blood Glucose (mg/dL)	253.5 $\pm$ 54.83	253.6 $\pm$ 61.90	253.5 $\pm$ 58.51

- Efficacy Results
- Change in FBG from Baseline
- At Weeks 6, 12 and 16, patients in both treatment arms demonstrated significant improvement in FBG from Baseline ($p<0.0001$ each).



- Proportion of Patients Achieving FBG <115 mg/dL at Week 16
- At Week 16, Test arm had higher proportion of patients achieving FBG < 115 mg/dL (Test vs Comparator: 27.8% vs 21.2%).

- Proportion of Patients Achieving both FBG < 115 mg/dL and HbA1c < 7% at Week 16
- In Test arm versus Comparator arm, the proportion of patients achieving both, target HbA1c < 7% and target FBG < 115 mg/dL at Week 6 and 12 was 17.4% vs 16.3% and 8.4% and 4.5%, respectively.
 - In Test arm, the proportion of patients achieving both, target HbA1c < 7% and target FBG < 115 mg/dL at Week 16 was statistically significant as compared to Comparator (Test vs Comparator: 14.2% vs 6.4%, $p=0.0123$).



- Safety Results
- Adverse Events – Overall Summary During the 16 Week Double Blind Treatment Period
- Safety profile was comparable between both arms, and the treatments were safe and well-tolerated.
 - Overall 67 TEAEs were reported in 55 (14.0%) patients; of which 31 TEAEs were reported in 25 (13.2%) patients in Test arm and 36 TEAEs were reported in 30 (14.9%) patients in Comparator arm.
 - 39 TEAEs (19 in Test arm and 20 in Comparator arm) were reported as ‘unlikely’ related to study products; 25 TEAEs (11 in Test arm and 14 in Comparator arm) were reported as ‘probable/likely’ related to the study products; and 3 TEAEs (1 in Test arm and 2 in Comparator arm) were reported as ‘possible’ related to the study products by investigator.
 - Action taken was “dose not changed” or “not applicable” for all the TEAEs in the study.
 - No SAE reported and none of the patient required dose modification.
 - The outcome of all reported TEAEs was resolved at the end of study.
 - None of the patient required rescue medications during the study.

Treatment Emergent Adverse Events by Preferred Term – During the 16 Week Double Blind Treatment Period

System Organ Class Preferred Term	Test group (N=190)		Comparator group (N=202)		Overall (N=392)	
	Patients [n (%)]	Events (E)	Patients [n (%)]	Events (E)	Patients [n (%)]	Events (E)
Total TEAEs	25 (13.2%)	31	30 (14.9%)	36	55 (14.0%)	67
Abdominal pain upper	1 (0.5%)	1	3 (1.5%)	3	4 (1.0%)	4
Constipation	0	0	1 (0.5%)	1	1 (0.3%)	1
Diarrhea	3 (1.6%)	3	5 (2.5%)	6	8 (2.0%)	9
Gastritis	0	0	2 (1.0%)	2	2 (0.5%)	2
Nausea	0	0	6 (3.0%)	6	6 (1.5%)	6
Vomiting	4 (2.1%)	5	4 (2.0%)	4	8 (2.0%)	9
Pyrexia	3 (1.6%)	3	5 (2.5%)	6	8 (2.0%)	9
Nasopharyngitis	1 (0.5%)	1	1 (0.5%)	1	2 (0.5%)	2
Hypoglycemia	3 (1.6%)	5	1 (0.5%)	2	4 (1.0%)	7
Arthralgia	1 (0.5%)	1	0	0	1 (0.3%)	1
Back pain	0	0	1 (0.5%)	1	1 (0.3%)	1
Myalgia	4 (2.1%)	4	0	0	4 (1.0%)	4
Headache	6 (3.2%)	6	3 (1.5%)	3	9 (2.3%)	9
Cough	1 (0.5%)	1	1 (0.5%)	1	2 (0.5%)	2
Rhinorrhea	1 (0.5%)	1	0	0	1 (0.3%)	1

- DISCUSSION AND CONCLUSION
- A statistically significant reduction in FBG was observed from Baseline to end of Week 16 in both the treatment arms.
 - Significantly more proportion of patients achieved target HbA1c of < 7% and target FBG of < 115 mg/dL with FDC of Sitagliptin+Glimepiride+Metformin at Week 16 compared to co-administration of Metformin and high dose Glimepiride.
 - Thus, the addition of Sitagliptin to Metformin and a lower dose of Glimepiride is superior in controlling glycemic parameters as compared to the co-administration of higher dose of Glimepiride with Metformin.
 - The hazard of vascular complications increases with increasing FBG level and hence control of FBG along with HBA1c control is required to prevent vascular complications.
 - Safety profile was comparable between both the arms, and both the study treatment were safe and well-tolerated.

Abbreviations:
BID: Twice daily; D: Day; E: Number of events; EOS: End of study; FBG: Fasting blood glucose, HbA1c: Glycated hemoglobin; N: Total number of patients; n: number of patients; PPBG: Postprandial blood glucose; SAE: Serious adverse events; SD: Standard deviation; T2DM: Type 2 diabetes mellitus; TEAEs: Treatment emergent adverse events; V: Visit; Wk: Week;

References:
1. Radhika G et al. Metabolism 2009 May;58(5):675-681.
2. Lazarus G et al. Diabetes Res Clin Pract 2021 Jan;171.
3. Makkar BM et al. Int J Diabetes Dev Ctries 2022 Oct;42 (Suppl 1):S1-S143.